Thinking 3D Object Aesthetics Assessment: Employing Hairstyle Aesthetics Assessment as An Exemplification

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Abstract

While traditional Image Aesthetics Assessment (IAA) relies heavily on 2D images, 3D Object Aesthetics Assessment (3D-OAA) remains underexplored due to challenges in modeling spatial geometry and reconciling contradictory multi-view aesthetic perceptions. To bridge this gap, we formulate a multi-view, 3D-aware assessment paradigm that is compatible with existing 2D IAA frameworks and comprises multi-view information embedding and adaptive view-weight allocation. We instantiate and evaluate this paradigm in Hairstyle Aesthetics Assessment (HAA), a texture-sensitive domain in which appearance varies across viewpoints, by introducing the Hairstyle Aesthetics Assessment Network (HAANet). On the one hand, HAANet evaluates hairstyle aesthetics by embedding multi-view portrait features and dynamically weighting their contributions. On the other hand, HAANet employs a two-stage multi-task learning strategy to extract specialized features and utilizes symmetric attention to capture aesthetic compatibility. Furthermore, we construct HAA10K, a comprehensive dataset of over 10,000 expert-annotated portrait images. On HAA10K, HAANet outperforms nine evaluated methods across all metrics, providing an initial benchmark for multi-view HAA and evidence for its use as an aesthetic signal in the evaluated downstream tasks. All resources will be available at here.

CCS Concepts

• **Computing methodologies** → **Computer vision**.

Keywords

Image Aesthetics Assessment, 3D Object Aesthetics Assessment, Hairstyle Aesthetics Assessment

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1 Introduction

Visual aesthetics is inherently a multi-dimensional and spatially complex perception. While Image Aesthetics Assessment (IAA) has advanced substantially, its reliance on a single 2D projection limits its representation of viewpoint-dependent appearance. A single projection can omit geometric, contextual, and structural cues that contribute to an object's aesthetic appeal. To bridge this gap, we formalize 3D Object Aesthetics Assessment (3D-OAA) as a multi-view, 3D-aware extension of IAA. It represents viewpoint-dependent appearance and spatial cues without claiming to recover complete 3D geometry. This shift reveals two fundamental questions:

Q1: What is the optimal representational format for 3D-OAA? Native 3D formats (e.g., meshes, point clouds, or neural radiance fields) preserve continuous geometry and topology, but current representations may lose fine-grained texture and require additional processing. Multi-view images provide a tractable, texture-preserving surrogate for HAA by sampling viewpoint-dependent appearance with standard 2D architectures. This representation does not recover continuous 3D topology and should therefore be interpreted as a 3D-aware approximation rather than a replacement for native 3D data.

Q2: How to reconcile asymmetric aesthetic perceptions across diverse viewpoints? In 3D-OAA, diverse viewpoints may yield distinct aesthetic interpretations, necessitating a robust framework to reconcile these differences. In a spatial context, different viewing angles capture drastically heterogeneous semantic information (e.g., a frontal view dictates compatibility with facial features, whereas a back view highlights volume and structural texture). Therefore, a robust 3D-OAA framework should transcend naive feature aggregation. It necessitates a dynamic, data-driven mechanism that not only embeds viewpoint-specific geometric priors

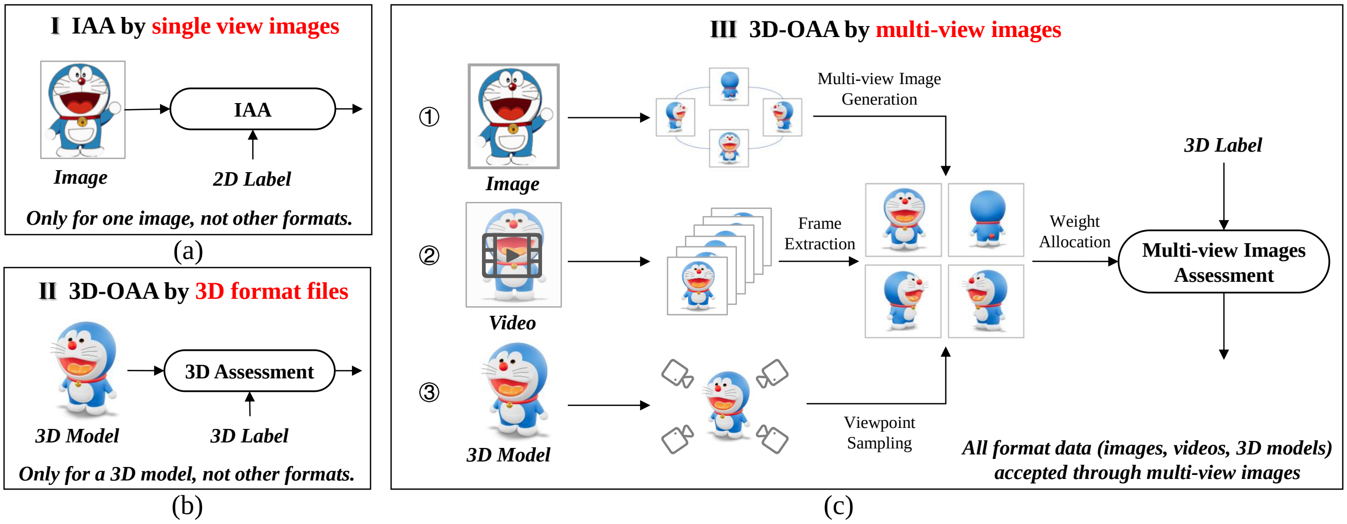


Figure 1: Conceptual comparison of existing paradigms and our proposed 3D Object Aesthetics Assessment (3D-OAA). Unlike 2D Image Aesthetics Assessment (IAA), which discards spatial geometry, or native 3D formats that suffer from texture degradation, our multi-view formulation provides a texture-preserving, 3D-aware representation of viewpoint-dependent aesthetic cues without recovering continuous 3D topology.

but also adaptively weights the contributions of each perspective based on its discriminative aesthetic value.

To ground this conceptual framework in a texture-sensitive scenario, we use Hairstyle Aesthetics Assessment (HAA) as a representative task. Hairstyles are intrinsically 3D structures whose appearance depends on multi-angle harmony and compatibility with facial morphology, making HAA suitable for evaluating the proposed paradigm. HAANet is a domain-specific instantiation with face and hair encoders; applying the paradigm to other objects would require task-specific encoders and separate validation. Our contributions are summarized as follows:

- We formulate a multi-view, 3D-aware assessment paradigm (Fig. 1) structured around multi-view information embedding and adaptive view-weight allocation.
- We evaluate the paradigm on the multi-view HAA task and design a specialized architecture, HAANet. We also construct HAA10K, a large-scale dataset of over 10,000 high-resolution portrait images featuring expert-annotated multi-view metadata, curated via a Human-AI collaborative pipeline.
- HAANet achieves state-of-the-art performance among the evaluated methods and provides an initial benchmark for multi-view HAA. The downstream user studies further examine its use as an aesthetic signal for image generation and hairstyle editing.

2 Related Work

Image Aesthetics Assessment. Traditional IAA has predominantly focused on quantifying visual appeal from a singular, fixed 2D perspective. While early methods relied on handcrafted features [6, 29, 30] and recent advancements have leveraged deep learning architectures, such as CNNs [14], Transformers [10, 13, 22, 38] and Multimodal Large Language Models (MLLMs) [16, 23, 31, 41, 44],to extract nuanced 2D features, these approaches share a fundamental limitation: they evaluate aesthetics based on flat, unidirectional projections. This narrow paradigm inherently struggles to capture the intricate, multi-dimensional nature of an object's aesthetic appeal. A truly comprehensive assessment requires evaluating an object from multiple viewpoints to account for spatial geometry and angle-dependent visual variations. Because current 2D IAA frameworks are ill-equipped to reconcile potentially contradictory aesthetic perceptions across different views, 3D-OAA remains significantly underexplored. Existing 3D research remains sparse, primarily focusing on evaluating the geometric quality of reconstructed objects [8, 9] or optimizing scene parameters, such as camera viewpoints [39]. Consequently, a significant gap persists between current academic methodologies and the robust demands of practical 3D-OAA applications.

Portrait and Hairstyle Editing and Assessment. Human-centric aesthetics assessment, particularly for portraits, has emerged as a crucial and rapidly evolving research domain [7, 35, 45]. Driven by the demand for personalized visual experiences, portrait generation and editing techniques have matured considerably. Foundational models like StyleGAN [19, 21] have revolutionized general image synthesis, while specialized frameworks such as Barbershop [46] and HairFastGAN [33] leverage latent-space embeddings to significantly enhance the precision of fine-grained manipulation. Concurrently, 3D portrait reconstruction models, including PanoHead [1] and SphereHead [25], have unlocked new geometric perspectives, laying the groundwork for complex multi-view applications.

Despite these generative strides and the availability of high-quality datasets such as CelebA-HQ [18] and FFHQ [20], the assessment of edited content has lagged significantly behind. Existing evaluative works, such as Facial Attractiveness Prediction (FAP) [5, 11, 26], primarily focus on the original, 2D facial geometry

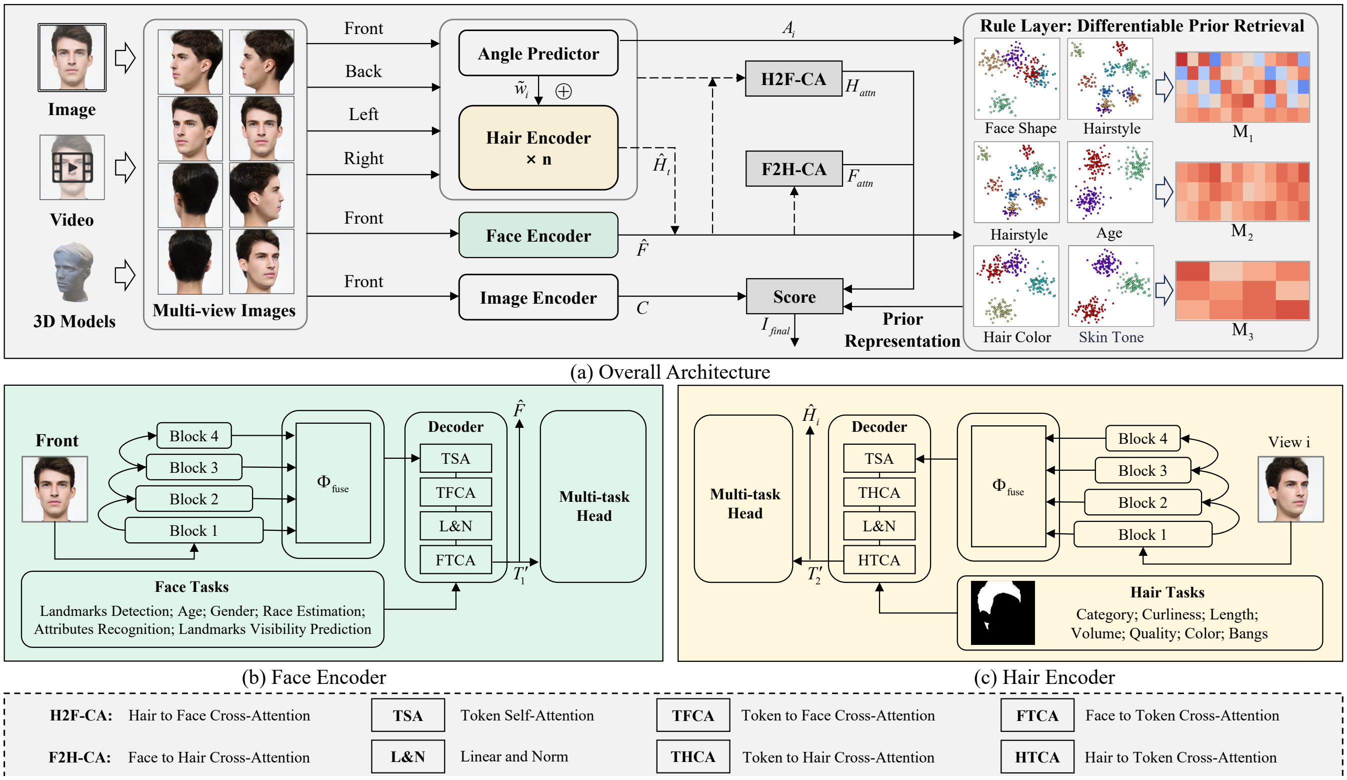


Figure 2: Architecture of the proposed HAANet. (a) The framework dynamically aggregates multi-view features via a viewpoint-aware gating mechanism, and injects structured aesthetic priors through a Differentiable Prior Retrieval module (Rule Layer). (b) The Face Encoder isolates frontal facial semantics via multi-task learning. (c) The Hair Encoder leverages bidirectional cross-attention to capture structural hair patterns, culminating in a unified aesthetic score.

and struggle to quantify the holistic impact of modified features. More importantly, when extending these generative techniques to multi-view and 3D scenarios, the field lacks dedicated, quantitative algorithms to accurately assess the aesthetic quality and geometric consistency of generated results.

This deficiency is particularly evident in hairstyle editing tasks, where current evaluation methodologies are fundamentally inadequate for aesthetic scoring, suffering from both objective and subjective limitations. Objective metrics, such as Identity Similarity (IDS) [17], measure algorithmic fidelity rather than personalized aesthetic impact. Similarly, structural metrics like PSNR and SSIM predominantly evaluate non-hair regions, inadvertently ignoring critical aesthetic factors such as hair texture and gloss, whereas Average Color Difference (ACD) captures only isolated chromatic shifts. Conversely, subjective user studies are notoriously costly, time-consuming, and prone to high variance caused by ambiguous interpretations of "naturalness" or "preservation" [40]. Furthermore, rigid manual annotations fail to adapt dynamically to novel or diverse hairstyles. Ultimately, traditional editing metrics focus on pixel-level fidelity or isolated attributes, making them fundamentally distinct from and insufficient for HAA.

3 Architecture of HAANet

Overall Architecture. Given the unique characteristics of HAA compared to traditional IAA tasks, we propose HAANet. As shown in Fig. 2, HAANet accepts multi-view portrait images (which can also be derived from other modalities, such as videos or 3D models) as input. The framework consists of three main extractors: the Hair Encoder extracts hairstyle features across multiple views, the Face Encoder processes frontal facial features, and the Image Encoder (based on ConvNeXt [28]) captures global context to provide a macro-level aesthetic reference.

Within the network, multi-view data is aggregated through a dynamic, viewpoint-aware fusion mechanism, and multi-modal features are aligned using synergistic bilateral cross-attention. Hair prior knowledge is explicitly injected via a Differentiable Prior Retrieval mechanism, which generates aesthetic guidance coefficients based on semantic prototyping and learnable bilinear compatibility matrices. The model is trained in a two-stage paradigm: 1) the encoders are independently pre-trained via multi-task learning to extract domain-specific features; 2) the pre-trained encoders are integrated with the Rule Layer for the final score regression.

3.1 Viewpoint-Aware Fusion

Multi-view Fusion in Hairstyle. When attempting multi-view fusion for hairstyle analysis, challenges such as inconsistent feature representations across views and the difficulty of integrating angle-specific details often undermine performance. To tackle this, we leverage an Angle Predictor to explicitly extract viewpoint information, which is then used as geometric positional encoding for the weighted fusion of multi-view features.

Each training sample consists of 12 multi-view images of a person's hair. From these, we select four images corresponding to the front, left, right, and back views, denoted as $I_1, I_2, I_3, I_4 \in \mathbb{R}^{H\times W\times 3}$. Each image I_i is processed separately by the Hair Encoder. Let S_{ij} represent the feature map extracted by the j-th block of the Hair Encoder for the i-th view image I_i. The combined hierarchical hair feature H_i for a specific view is formulated as:

$$H_i = \Phi_{\text{fuse}}\left(\text{Concat}_{j=1}^{L}(S_{ij})\right), \tag{1}$$

where Concat denotes channel-wise concatenation, L is the total number of blocks, and $\Phi_{\text{fuse}}(\cdot)$ represents the non-linear projection mapping. Next, learnable task-specific tokens T_2 are introduced for hair-related tasks (e.g., hair attribute and mask prediction).

Let $\text{Attn}(Q, K, V) = \text{softmax}\left(QK^\top/\sqrt{d_k}\right)V$ denote the standard scaled dot-product attention, where d_k is the dimension of the key vectors. These task tokens are first updated using Token Self-Attention (TSA):

$$T_2' = \text{Attn}(T_2W_Q^t, T_2W_K^t, T_2W_V^t), \tag{2}$$

where W_Q^t, W_K^t, W_V^t are the learned linear projections. To establish bidirectional feature interaction as depicted in Fig. 2(c), the updated task tokens T_2' process the hair features H_i via Token-to-Hair Cross-Attention (THCA):

$$\hat{H}_i = \text{Attn}(H_iW_Q^h, T_2'W_K^h, T_2'W_V^h), \tag{3}$$

where W_Q^h, W_K^h, W_V^h represent the corresponding learnable linear projection matrices for the THCA module. Symmetrically, a Hair-to-Token Cross-Attention (HTCA) module is employed to update the task tokens using the hair features, ensuring that the task representations effectively capture structural hair patterns.

To address the inherent informational asymmetry across different perspectives, we design a data-driven gating mechanism. The normalized weight $\tilde{w}_i$ for each view is conditionally computed based on an energy function $\mathcal{E}_{\text{gate}}$ over the visual feature and its geometric viewpoint:

$$\tilde{w}_i = \frac{\exp\left(\mathcal{E}_{\text{gate}}([\hat{H}_i \parallel A_i])\right)}{\sum_{v=1}^{4}\exp\left(\mathcal{E}_{\text{gate}}([\hat{H}_v \parallel A_v])\right)}, \tag{4}$$

where $\mathcal{E}_{\text{gate}}$ is an energy function parameterized by a lightweight multi-layer perceptron (MLP), $[\cdot \parallel \cdot]$ denotes feature concatenation, and A_i denotes the angle-specific feature generated by the *Angle Predictor* (optimized via cross-entropy loss), acting as a latent surrogate for explicit 3D camera extrinsics. This allocation allows the network to assign higher weights to viewpoints with greater discriminative value.

Finally, the multi-view hair features are aggregated via a convex combination conditionally scaled by the gating weights:

$$\hat{H}_t = \sum_{i=1}^{4}\tilde{w}_i \cdot \left(\alpha\hat{H}_i + (1-\alpha)A_i\right), \tag{5}$$

where the subscript t indicates the aggregated multi-view representation, and α acts as a modality-balancing coefficient between visual texture and spatial geometry.

Synergistic Bilateral Cross-Attention. Existing models often treat face and hairstyle features independently, neglecting their high correlation in determining overall portrait aesthetics. To capture their complex interactions and model their joint aesthetic distribution, we design a Synergistic Bilateral Cross-Attention mechanism.

Specifically, the face feature $\hat{F}$ and the aggregated global hairstyle feature $\hat{H}_t$ are mutually processed. The resulting attention features, F_{attn} and H_{attn}, represent the interactively fused representations:

$$F_{\text{attn}} = \text{Attn}(\hat{F}W_Q^f, \hat{H}_tW_K^f, \hat{H}_tW_V^f), \tag{6}$$

$$H_{\text{attn}} = \text{Attn}(\hat{H}_tW_Q^c, \hat{F}W_K^c, \hat{F}W_V^c), \tag{7}$$

where W_Q^f, W_K^f, W_V^f and W_Q^c, W_K^c, W_V^c are the specific learnable linear projection matrices for the Face-to-Hair (F2H-CA) and Hair-to-Face (H2F-CA) cross-attention branches, respectively.

To further regularize the local interactions, the global feature C, extracted from the frontal input image I_{front} by the Image Encoder, is retained. This global feature helps the network achieve an optimal balance between holistic visual context and localized fine-grained details.

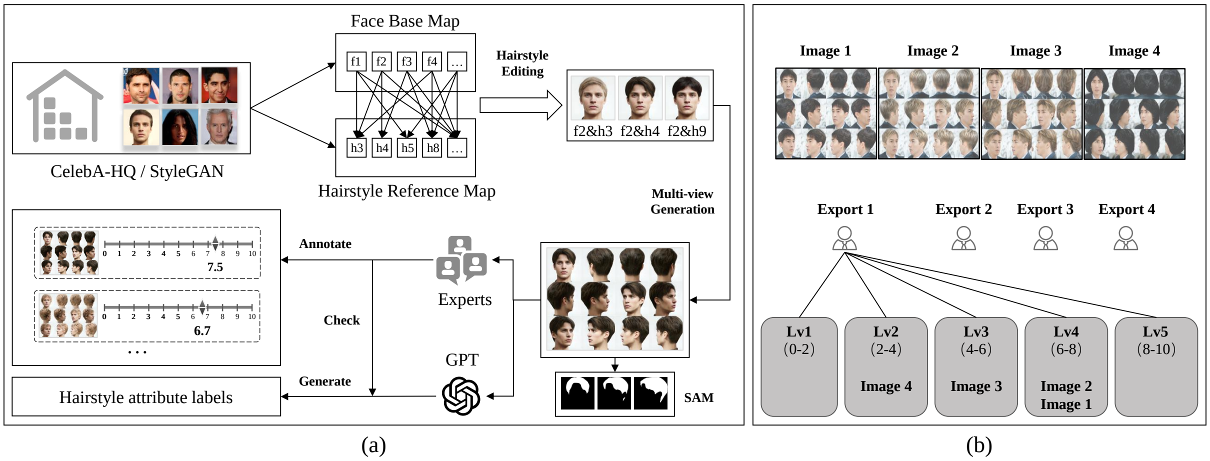
3.2 Encoder Training and Prior Injection

Multi-task Training Process of Encoder. Developing robust encoders that are sensitive to semantic regions (e.g., face and hair) is challenging. We design multiple sub-tasks to force the encoders to focus strictly on their respective regions, disentangling local semantics from the global aesthetic assessment.

1) Face Encoder. Inspired by *FaceXFormer* [32], we utilize a transformer-based architecture. The initial facial token T_1 is first processed via self-attention to produce T_1'. Through subsequent Token-to-Face Cross-Attention (TFCA) and Face-to-Token Cross-Attention (FTCA) interactions, it eventually yields the final facial feature $\hat{F}$. The multi-task learning objectives here include face parsing, landmark detection, head pose estimation, attribute recognition, age/gender/race estimation, and landmarks visibility prediction, which are jointly optimized via a composite loss function applied to specific multi-task heads.

2) Hair Encoder. Similarly, we optimize the Hair Encoder using a combined loss $L = \sum_{i=1}^{m}\lambda_iL_i$, where m represents the number of sub-tasks, λ_i is the loss weight, and L_i denotes the loss function for the i-th specific sub-task. The sub-tasks for the hair region involve predicting the hair mask, category, curliness, length, volume, quality, color, and bangs.

Differentiable Prior Retrieval. Aesthetic assessment heavily relies on human prior knowledge (e.g., the harmony between face shape and hairstyle), which is notoriously difficult to model implicitly. To bridge this gap, we design a Rule Layer that explicitly introduces structured prior knowledge via semantic prototyping and bilinear queries.



	Lv1	Lv2	Lv3	Lv4	Lv5
Export 1		Image 4	Image 3	Image 2<Image 1	
Export 2		Image 4<Image 3		Image 2<Image 1	
Export 3		Image 4	Image 3	Image 1<Image 2	
Export 4				Image 3<Image 2<Image 1	Image 4

	Image 1	Image 2	Image 3	Image 4
Export 1	7.33	6.67	5	3
Export 2	7.33	6.67	3.33	2.67
Export 3	6.67	7.33	5	3
Export 4	7.5	7	6.5	9 ×

Image 1	Image 2
7.21	6.92
Image 3	**Image 4**
4.96	2.89

(c)

Figure 3: The dataset construction pipeline. (a) Generation and manual quality screening of diverse multi-view face-hairstyle pairings. (b) The expert-driven contrastive annotation protocol is designed to mitigate subjective bias across five macroscopic tiers. (c) An example of translating discrete intra-tier rankings into continuous aesthetic scores via linear interpolation.

Before the final score regression, we compute three aesthetic subscores: face-shape-hairstyle, hairstyle-age, and hair-skin-color. We construct semantic prototype sets $\mathcal{P}_{\text{face}} = \{\mu_m^{\text{face}}\}_{m=1}^{K_f}$ and $\mathcal{P}_{\text{hair}} = \{\mu_n^{\text{hair}}\}_{n=1}^{K_h}$ by applying K-Means clustering over the latent feature space of the entire training set, with $K_f = 5$ and $K_h = 12$.

During forward propagation, we compute the soft assignment probabilities over the probability simplex using temperature-scaled cosine similarity. Let $\mathbf{p}_{\text{face}} \in \mathbb{R}^{K_f}$ and $\mathbf{p}_{\text{hair}} \in \mathbb{R}^{K_h}$ be the continuous probability vectors, where the m-th and n-th elements are respectively given by:

$$\mathbf{p}_{\text{face}}^{(m)} = \frac{\exp\left(\beta \cdot \cos\left(\hat{F}, \mu_m^{\text{face}}\right)\right)}{\sum_{k=1}^{K_f} \exp\left(\beta \cdot \cos\left(\hat{F}, \mu_k^{\text{face}}\right)\right)}, \tag{8}$$

$$\mathbf{p}_{\text{hair}}^{(n)} = \frac{\exp\left(\beta \cdot \cos\left(\hat{H}_t, \mu_n^{\text{hair}}\right)\right)}{\sum_{k=1}^{K_h} \exp\left(\beta \cdot \cos\left(\hat{H}_t, \mu_k^{\text{hair}}\right)\right)}. \tag{9}$$

Here, β acts as a temperature scaling parameter controlling the sharpness of the probability distribution, and $\cos(\cdot,\cdot)$ denotes the cosine similarity function. The probability vectors define soft weights over all prototypes, so each sample is represented as a mixture rather than assigned to a single centroid. The face-to-hairstyle compatibility score R_1 is then computed as a bilinear form over the probability simplexes:

$$R_1 = \mathbf{p}_{\text{face}}^{\top} \mathbf{M}_1 \mathbf{p}_{\text{hair}}, \tag{10}$$

where $\mathbf{M}_1 \in \mathbb{R}^{K_f \times K_h}$ is an expert-initialized learnable compatibility matrix. Professional hairstylists assign compatibility scores to semantic pairs, including face shape–hairstyle, hairstyle–age, and hair color–skin color. After aggregation and normalization, these scores initialize $\mathbf{M}_1$, $\mathbf{M}_2$, and $\mathbf{M}_3$, respectively; the matrices are not fixed lookup tables. They are queried to produce R_1, R_2, and R_3, forming the concatenated prior representation $R = [R_1, R_2, R_3]$.

Dynamic Updating Mechanism: During score-regression training, $\mathbf{M}_1$, $\mathbf{M}_2$, and $\mathbf{M}_3$ remain learnable and are optimized with the regularization term. The semantic prototypes are refreshed every 10 epochs to track the evolving feature space. This differentiable retrieval mechanism uses expert knowledge as a structured initialization while allowing data-driven adaptation.

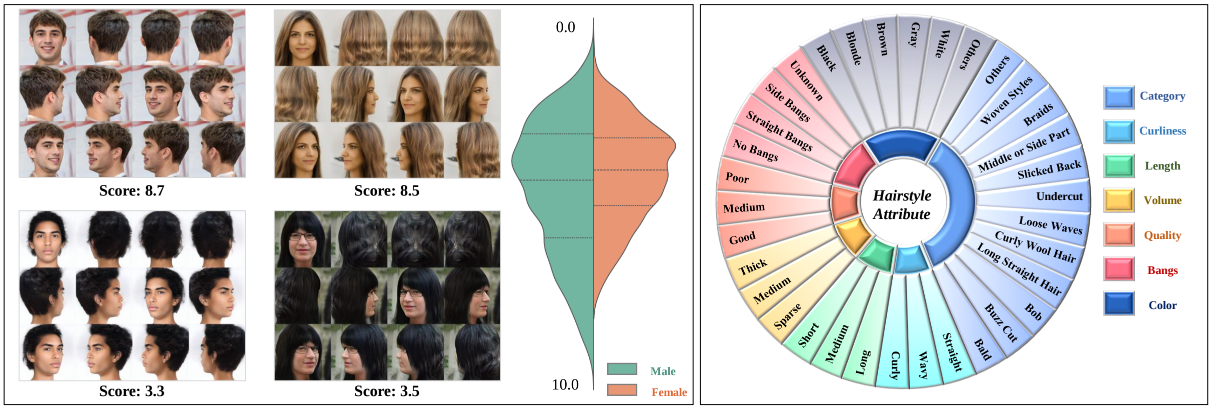
Finally, the ultimate aesthetic representation I_{final} is constructed by integrating the interactively fused visual features:

$$I_{\text{final}} = \left[\Phi_f(F_{\text{attn}} \,\|\, C) \,\middle\|\, \Phi_h(H_{\text{attn}} \,\|\, C) \,\middle\|\, R\right], \tag{11}$$

where $\Phi_f(\cdot)$ and $\Phi_h(\cdot)$ denote non-linear projection mappings used to map the concatenated local and global features into a unified latent space. This representation I_{final} is subsequently fed into an MLP head to output the final aesthetic score.

4 The HAA10K Dataset

To evaluate the proposed 3D-OAA paradigm, we introduce HAA10K, a large-scale, multi-view benchmark curated for this task. HAA10K comprises over 10,000 high-resolution portrait images with intra-individual hairstyle variations and multi-view observations. Its construction pipeline is designed to support data diversity, annotation consistency, and visual quality, as illustrated in Fig. 3.



(a) (b)

Figure 4: Visual samples and attribute of the HAA10K dataset. (a) Representative multi-view images with continuous aesthetic scores and the overall score distribution segmented by gender. (b) The comprehensive hairstyle attribute taxonomy is utilized to supervise the multi-task pre-training of the Hair Encoder.

4.1 Data Collection and Quality Control

Constructing a high-quality multi-view dataset necessitates a delicate balance between morphological diversity and cross-view consistency. We achieve this through a highly controlled, three-stage generation and screening pipeline (Fig. 3(a)).

First, to establish a morphologically diverse foundation, we construct a *Face Base Map* by curating 1,000 high-fidelity portraits from CelebA-HQ. We also incorporate StyleGAN-synthesized pseudo-portraits to broaden the distributions of demographics and facial morphology under standardized poses and lighting. Second, we establish a *Hairstyle Reference Map* comprising 200 representative hairstyles organized into 20 typologies (e.g., long/short, curly/straight, and dyed variations). Using HairFastGAN and HairCLIP, each identity in the *Face Base Map* is combined with 20 randomly sampled hairstyles to produce diverse face-hairstyle pairings. Finally, we use the 3D-aware generators PanoHead and SphereHead to synthesize 12 discrete viewpoints per customized portrait.

Quality Assurance against Generative Bias: We explicitly acknowledge that synthetic pipelines may introduce structural artifacts or inductive biases. To mitigate this stringently, we instituted a massive manual screening protocol. Exhaustive human-in-the-loop inspections were conducted to discard samples exhibiting texture degradation, geometric distortions, or multi-view inconsistencies. Only images demonstrating photorealistic fidelity and strict multi-angle alignment were retained, effectively suppressing the distribution gap and generative defects.

4.2 Human-AI Collaborative Annotation

To furnish the dataset with rich semantic labels, we employ a highly efficient Human-AI collaborative workflow.

1) Structural Masks: Accurate parsing masks for semantic regions are automatically extracted using the Segment Anything Model (SAM) and BiSeNet, providing reliable spatial priors.

2) Fine-grained Attributes: As shown in Fig. 4(b), attributes encompassing category, curliness, length, volume, quality, color, and bangs are initially predicted using Vision-Language Models (e.g., GPT-4V). Crucially, to ensure absolute semantic correctness, every machine-generated label undergoes strict manual verification and correction by human annotators.

4.3 Expert-Driven Contrastive Aesthetic Scoring

A central challenge in aesthetics assessment is controlling annotator subjectivity. To address the "Special Relativity" of IAA [43], we use an expert contrastive ranking protocol (Fig. 3(b)).

We recruited 20 professional hairstylists for aesthetic scoring. Experts compare 10–20 images of the *same individual* with different hairstyles, which reduces confounding from differences in facial attractiveness. All samples are first assigned to five shared global tiers (*Lv1: Bad* to *Lv5: Excellent*) mapped to a common 0–10 scale. These common tier anchors support cross-individual score comparability. Within each tier, experts then conduct fine-grained relative ranking for samples of the same identity.

Continuous scores are computed by linear interpolation only within the expert-assigned tier, preserving the shared global tier label. For instance, if two images fall into *Lv4* (6–8 points), their intra-tier order maps them to 6.67 and 7.33. An isolated image within a tier is assigned the median value (e.g., *Lv2* receives 3.0 points). Before computing the final Mean Opinion Score (MOS), robust statistical deviation thresholds are applied to filter annotator outliers. The resulting annotations yield Kendall's $W = 0.72$, an intraclass correlation coefficient of 0.78, and an outlier rate of 4.6%.

This protocol combines shared global anchors with ranking within each identity to produce comparable continuous scores while reducing identity-related confounding.

5 Experiments

5.1 Experimental Settings

Evaluation Metrics. We adopt three standard metrics to evaluate performance: Spearman Rank Correlation Coefficient (SRCC, $\mathcal{S}$), Linear Correlation Coefficient (LCC, $\mathcal{L}$), and Mean Squared Error

Table 1: Comparison of HAANet with the evaluated models on HAA10K.

Metric	IAA				FAP		MLLM		Ours
	EAT	TANet	NIMA	BLG-PIAA	CNN-ER	MEBeauty	Qwen2.5-VL	Q-Align	HAANet
$\mathcal{L}\uparrow$	0.617	0.647	0.665	0.650	0.529	0.632	0.437	0.547	**0.781**
$\mathcal{S}\uparrow$	0.552	0.603	0.611	0.598	0.465	0.582	0.388	0.496	**0.742**
$\mathcal{M}\downarrow$	1.380	1.228	1.134	1.196	1.648	1.180	2.006	1.634	**0.824**
$\mathcal{A}\uparrow$	0.536	0.566	0.588	0.576	0.482	0.559	0.497	0.522	**0.682**

(MSE, $\mathcal{M}$). Additionally, we introduce Accuracy (Acc, $\mathcal{A}$) to measure the exact classification match across the five discrete score levels (Lv1-Lv5). To ensure statistical robustness, all reported results are averages from a 5-fold cross-validation. The network is optimized using AdamW on a single NVIDIA RTX 4090 GPU.

Benchmark Models. To the best of our knowledge, there are no publicly available datasets or models specifically tailored for HAA. To ensure a fair comparison, we evaluate the performance of 9 SOTA models from the IAA and FAP domains on our HAA10K dataset. These benchmarks provide a comprehensive reference for assessing the effectiveness of our proposed approach.

Training Process. HAANet is trained in two stages. In the first stage, we use a multi-task approach to pre-train the feature extractors for both hairstyles and faces, utilizing our in-house Hairstyle Attribute Data (Fig. 4(b)) for hairstyles and various publicly available datasets for faces. The second stage focuses on training the score-fitting model on the HAA10K dataset, with an 80% training and 20% test split. The training loss consists of a score-fitting loss, an angle-prediction loss, and a regularization loss for the rule matrices, which are initialized using expert annotations.

Input-to-Multi-view Transformation. Our framework supports flexible input modalities, including single images, video sequences, and 3D models. These inputs are transformed into standardized multi-view representations [1, 25] for unified aesthetics assessment. Specifically, images are converted into multiple views via multi-view image generation; video sequences are processed by frame extraction and viewpoint sampling; and 3D models are sampled at various viewpoints to generate the necessary multi-view images. The reported HAANet inference speed measures the forward pass on standardized multi-view inputs. When a single image requires view synthesis, optional PanoHead/SphereHead preprocessing takes approximately 2.6 s per sample; this cost is reported separately from HAANet inference.

5.2 Performance Comparison

Performance on HAA10K. We evaluate the performance of several SOTA models from the IAA and FAP domains on our HAA10K dataset, including EAT [13], TANet [14], NIMA [37], and BLG-PIAA [45] from the IAA domain, and CNN-ER [4] and MEBeauty [24] from the FAP domain. The results, in Table 1, demonstrate that our method outperforms all baseline models on the established metrics, using the average of all views for single-image input models. We also compare fine-tuned MLLMs, Qwen2.5-VL [2] and Q-Align [42], which currently perform suboptimally on aesthetic scoring tasks due to limitations in the dataset and the medium-score prediction bias. Our experiments indicate that models tailored for hairstyle aesthetics yield better results than general aesthetics models, particularly for FAP models, showing notable improvements in $\mathcal{M}$ and $\mathcal{A}$ metrics. This highlights that our approach, which focuses on the aesthetic interaction between the face and hairstyle, captures the unique charm of hairstyles beyond the beauty of the face alone.

Zero-shot Generalization on Real-world Scenarios. A potential concern with models trained on synthesized multi-view datasets is the risk of overfitting to the inductive biases of the generative pipeline. To rigorously validate that HAANet learns genuine 3D aesthetic representations rather than generative artifacts, we construct a challenging **real-world evaluation subset (HAA-Real-100)**.

We manually captured and curated 100 wild videos/multi-view image arrays of real individuals, annotated by experts following the exact protocol used for HAA10K. In a strict Zero-shot setting, HAANet achieves an impressive SRCC of 0.721 ± 0.015 and an Accuracy of 55.4% ± 1.2%. Although there is a marginal performance drop compared to the synthetic test set due to the inevitable domain gap (e.g., wild lighting and background clutter), HAANet still outperforms the best fully-supervised 2D IAA baseline by over 12% in SRCC. This decisively proves the strong generalization capability of our multi-view framework to authentic real-world 3D objects.

5.3 Ablation Study

Robustness to View Sparsity and Angle Perturbations. To understand the impact of multi-view inputs and address potential real-world constraints where 360° views are unavailable, we ablate view sparsity during inference using the HAANet model, which is already fully trained on the HAA10K dataset. When reducing the test inputs to only 2 views (front and back), the model achieves an SRCC of 0.715 and an Acc of 0.674, demonstrating a graceful performance degradation rather than a catastrophic failure. When further restricted to a single frontal view, the SRCC and Acc drop to 0.439 and 0.498, respectively. Although there is a noticeable decrease compared to the full 4-view setting, the model still captures fundamental aesthetic cues, aided by the global context prior. Furthermore, we test the model's robustness to horizontal head rotations by introducing ±15° angular noise to the test inputs. The SRCC decrease is below 0.03, indicating that the Angle Predictor can compensate for moderate viewpoint shifts. However, severe viewpoint misalignment and heavy occlusion remain challenging and are not resolved by the current Angle Predictor.

Analysis of Data-Driven Gating Weights. We analyze the learned gating weights $\tilde{w}_i$ to validate our viewpoint-aware fusion. On the HAA10K test set, the front and back views consistently dominate, with average weights of 38.4% and 29.6%, compared to 16.2% and 15.8% for the left and right sides, respectively. This asymmetric

Table 2: Ablation study on backbone architectures for the Image Encoder.

Models	$\mathcal{L}\uparrow$	$\mathcal{S}\uparrow$	$\mathcal{M}\downarrow$	$\mathcal{A}\uparrow$
ViT	0.755	0.725	0.867	0.669
MobileNet-V3	0.757	0.712	0.868	0.670
SwinT	0.767	0.728	0.841	0.670
ResNet-50	0.777	0.736	0.838	0.680
ConvNeXt	**0.781**	**0.742**	**0.824**	**0.682**

Table 3: Performance and computational efficiency comparison across input modality paradigms for HAA on HAA10K. Inference FPS measures the network forward pass on standardized multi-view inputs and excludes optional view-generation preprocessing.

Paradigm	Param. (M) ↓	Inference (FPS) ↑	$\mathcal{S}\uparrow$	$\mathcal{A}\uparrow$
Native 3D Mesh [36]	158.3	12.5	0.631	0.594
Video-based [3]	345.1	6.2	0.702	0.651
HAANet	**42.5**	**45.0**	**0.742**	**0.682**

distribution is consistent with 3D aesthetic priors: the frontal view dictates crucial face-to-hairstyle compatibility, while the back view highlights volume and structural texture, suggesting that our model adaptively prioritizes discriminative angular features.

Role of the Rule Layer. We assess the impact of the Rule Layer, which incorporates prior knowledge of hairstyling. Removing this component results in decreased accuracy (SRCC=0.717, Acc=0.658), particularly in evaluating the compatibility between hairstyles and facial features. This highlights the importance of integrating domain-specific semantic prototypes, such as face-to-hairstyle compatibility, to improve the precision of aesthetic scores.

Image Encoder Backbone Selection. To extract the global context C, we evaluate five mainstream backbones: ConvNeXt [28], ResNet-50 [12], ViT [10], Swin [27], and MobileNet-V3 [15]. As shown in Table 2, ConvNeXt achieves the strongest performance among the evaluated backbones. This result supports its use as the global-context encoder alongside the fine-grained features extracted by the multi-view branches.

5.4 Comparison with 3D and Video Paradigms

To compare the multi-view image representation with alternative formats for HAA, we evaluate HAANet against native 3D mesh and video-based approaches:

1) Native 3D Mesh. We construct a baseline using DiffLocks [36] to extract geometric features directly from the 3D hair meshes derived during HAA10K construction. As shown in Table 3, the mesh-based approach yields lower SRCC and Acc than HAANet in this HAA setting. One likely factor is that hairstyle aesthetics depend on fine-grained texture and material cues, including hair luster, color gradients, and individual strands, which can be degraded by the resolution and texturing of the evaluated meshes. These results support multi-view images as a texture-preserving surrogate for HAA; they do not imply that continuous geometry is unimportant or that multi-view images replace native 3D representations for all object categories.

2) Video-based. We treat the multi-view array (12 views) as a temporal sequence and use TimeSformer [3], a standard Video Transformer, to capture spatio-temporal features. The video-based approach achieves performance closer to HAANet but uses 345.1M parameters and runs at 6.2 FPS. Compared with TimeSformer, HAANet improves SRCC by 4.0%, uses 12.3% of the parameters, and is over 7× faster on the standardized multi-view inputs.

Table 4: User study results for generative downstream tasks guided by HAANet. The table reports preference rates for HAANet-guided image generation and hairstyle editing relative to their respective baselines.

Application Task	Baseline	HAANet-Guided	Pref. (Baseline)	Pref. (Ours) ↑
Image Generation	Default Seed	**Best-of-20**	23.5%	**76.5%**
Hairstyle Editing	Manual Tuning	**Latent Opt.**	18.8%	**81.2%**

5.5 Enhancing Generative Models with HAANet

To examine whether HAANet can provide an aesthetic signal for downstream generative tasks, we evaluate aesthetic-guided generation and aesthetic-driven editing (Table 4).

1) Aesthetic-Guided Image Generation. Text-to-image models [34] often lack explicit 3D aesthetic constraints. We utilize HAANet as a reward metric for Best-of-N (N=20) rejection sampling across 100 prompts. A double-blind user study (50 participants) shows our method is preferred 76.5% of the time over random-seed baselines, with users noting superior hair volume and facial compatibility.

2) Aesthetic-Driven Hairstyle Editing. In frameworks like Barbershop [46], natural blending typically requires tedious manual tuning. Leveraging HAANet's differentiability, we integrate its compatibility score (R_1) as an objective function to optimize generative latents via gradient descent. The resulting edits achieve an 81.2% preference rate relative to manual tuning for structural harmony and editing naturalness.

6 Conclusion

In this paper, we introduce a multi-view, 3D-aware assessment paradigm that uses viewpoint-aware fusion to represent complementary aesthetic cues across views. Within HAA, we instantiate this paradigm as HAANet, construct the HAA10K dataset, and establish multi-view benchmarks. HAANet achieves state-of-the-art performance among the evaluated methods and provides an aesthetic signal for the evaluated generative downstream tasks. The current evidence is limited to HAA. Multi-view images preserve texture and viewpoint-specific appearance but do not encode continuous 3D topology. Moreover, HAANet's face and hair encoders are domain-specific; extending the paradigm to other object categories requires task-specific encoder adaptation and independent validation. Severe view misalignment and heavy occlusion also remain open challenges.

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